

Highlights

Multimodal Medical Knowledge Fusion Framework: Integrating DeepSeek Language Models with Neo4j Knowledge Graphs for Advanced Medical Question Answering

Yanzhe Fan,Jing Han

- Proposed MMKF, a "Glass Box" neural-symbolic framework that enforces **Explicit Symbolic Grounding** to resolve the opacity and hallucination issues of medical LLMs.
- Achieved 92% accuracy and near-zero hallucinations using a **Safety Valve Mechanism** that dynamically aligns neural confidence with knowledge graph structural density.
- Leveraged **Unslow** optimization to reduce the computational overhead of fine-tuning DeepSeek models by 50%, democratizing access to high-performance medical AI.
- Solved the "Knowledge Cutoff" problem via **Unseen Entity Alignment**, enabling accurate reasoning about novel drugs (e.g., *Donanemab*) absent from the training data.
- Implemented a bidirectional alignment mechanism that synergizes the generative fluency of LLMs with the rigorous logical constraints of Neo4j graphs.

Multimodal Medical Knowledge Fusion Framework: Integrating DeepSeek Language Models with Neo4j Knowledge Graphs for Advanced Medical Question Answering

Dr. Yanzhe Fan^{a,1}, Dr. Jing Han^{a,*}

^aWuhan Business University, 816 Dongfeng Avenue, Wuhan Economic and Technological Development Zone, Wuhan, 430056, Hubei Province, China

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ABSTRACT

The deployment of Large Language Models (LLMs) in healthcare is severely constrained by their inherent “Black Box” nature and the associated risk of hallucinations. While emerging research in medical vision-language models attempts to mitigate this through visual grounding (e.g., GEMeX-RMCot), a comparable mechanism for **explicit symbolic grounding** in textual reasoning remains critical yet underexplored. This paper proposes the **Multimodal Medical Knowledge Fusion Framework (MMKF)**, a novel neural-symbolic architecture designed to resolve the interpretability crisis in medical QA. By strategically integrating DeepSeek LLMs with Neo4j graph databases, our framework anchors neural generation to verifiable knowledge graph paths, ensuring that every answer is supported by traceable structural evidence. The system architecture encompasses three innovative components: comprehensive medical knowledge graph construction, domain-adaptive fine-tuning of DeepSeek models via efficient Unsloth technology, and a bidirectional alignment mechanism between unstructured text and structured graphs. Extensive experimental validation across diverse medical specialties demonstrates that MMKF not only achieves substantial improvements in accuracy (92% vs. 89% baseline) but also significantly enhances the trustworthiness and auditability of medical AI. Furthermore, the implementation of Unsloth reduces computational resource requirements by approximately 50%, facilitating broader accessibility. This research provides a foundational paradigm for shifting medical AI from opaque inference to transparent, grounded reasoning.

1. Introduction

The healthcare industry is currently witnessing an unprecedented expansion in data generation, with growth rates substantially exceeding those observed in other sectors [1]. This expansive dataset encompasses diverse and complex information modalities, including molecular drug structures, clinical trial outcomes, disease phenotypes, and adverse drug reaction profiles. Collectively, these data points form the foundational knowledge base for contemporary medical research and clinical practice. However, managing the inherently complex, interconnected knowledge structures that characterize the medical domain poses significant challenges for traditional relational database systems. Graph databases, particularly Neo4j, have emerged as a superior paradigm, offering a more intuitive and computationally efficient method for organizing and retrieving these interconnected medical entities [2].

Despite the structural advantages of Knowledge Graphs (KGs) [3, 4], the recent surge in Artificial Intelligence has been dominated by Large Language Models (LLMs). LLMs have showcased exceptional capabilities in Natural Language Understanding (NLU) and Generation (NLG) [5]. However, the direct deployment of general-purpose LLMs in high-stakes medical domains is fraught with peril. The notorious “Black Box” nature of deep neural networks, coupled with the phenomenon of “hallucination”—where

models generate plausible-sounding but factually incorrect information—presents an unacceptable risk for clinical decision-making [6, 7]. In medicine, accuracy without interpretability is insufficient; a diagnosis or treatment recommendation must be auditable and grounded in established fact.

To mitigate the Black Box problem, the AI research community is increasingly turning towards “**Explicit Grounding**” mechanisms. In the realm of medical visual question answering (Med-VQA), emerging paradigms such as the GEMeX-RMCot framework [8] attempt to achieve **Visual Grounding** by anchoring reasoning steps to specific bounding boxes on medical images. While this visual transparency is crucial, a parallel and equally critical gap remains in the domain of textual and logical reasoning: the need for **Symbolic Grounding**. There is an urgent requirement for systems that can anchor neural text generation not just to pixels, but to verifiable, structured knowledge paths, thereby preventing semantic drift and logical fabrication.

To address this challenge, this paper proposes the **Multimodal Medical Knowledge Fusion Framework (MMKF)**, a novel neural-symbolic architecture specifically tailored to bridge the gap between neural fluency and symbolic rigour. The core innovation of MMKF lies in its ability to facilitate **Explicit Symbolic Grounding**. By efficiently fine-tuning the open-source DeepSeek LLM [9] using the Unsloth optimization library [10, 11] and integrating it with a comprehensive Neo4j-based medical knowledge graph, we create a system where answers are not merely generated, but

*Corresponding author.

✉ mashiroy4545456@gmail.com (Y. Fan); hjhust2014@163.com (J. Han)

ORCID(s):

¹First author.

retrieved and synthesized from traceable knowledge paths (Retrieval Augmented Generation) [12].

The primary contributions of this work are as follows:

1. **Neural-Symbolic Grounding Architecture:** We propose an end-to-end intelligent medical QA architecture that synergizes the generative power of DeepSeek with the structural integrity of Neo4j, establishing a paradigm for symbolically grounded medical reasoning.
2. **Verifiable Efficiency:** We validate the effectiveness of Unsloth optimization, demonstrating a reduction in computational resource requirements for domain-specific fine-tuning by approximately 50%, making high-performance medical AI more accessible and sustainable.
3. **Evidence-Based Performance:** Through rigorous evaluation, we demonstrate that symbolic grounding leads to substantial improvements in answer accuracy (92% vs. 89% baseline) and response relevance, proving that interpretability does not come at the cost of performance.
4. **Traceable Integration Methodology:** We explore innovative methods for integrating KGs with fine-tuned LLMs, offering a novel approach where model outputs can be audited against specific nodes and edges in the knowledge graph.
5. **Clinical Trust Framework:** Beyond technical metrics, we provide a concrete technical paradigm for building trustworthy, auditable LLM applications suitable for deployment in high-risk, specialized medical fields.

2. Related Work

The evolution of medical QA systems has transitioned from rule-based approaches to sophisticated neural-symbolic frameworks. This section reviews the critical components of this evolution, highlighting the gap that necessitates explicitly grounded reasoning.

2.1. Medical Knowledge Graphs as Structured Evidence

Medical Knowledge Graphs (MKGs) provide the structural backbone for symbolic reasoning in healthcare. By modeling entities (diseases, drugs, symptoms) and their interconnections (causes, treats, interacts-with), MKGs offer a deterministic and verifiable representation of medical facts [13]. Contemporary graphs, utilizing ontologies such as SNOMED-CT and RxNorm, contain millions of relations, supporting complex inferential hops that are difficult for unstructured models to perform reliably [14]. Unlike statistical models, MKGs provide definitive paths for reasoning, acting as the “gold standard” for symbolic grounding [4].

2.2. The Black Box Dilemma in Medical LLMs

While Large Language Models (LLMs) like Med-PaLM 2 have demonstrated expert-level performance on medical licensing exams [15], they remain fundamentally opaque. The “Black Box” nature of these models means their reasoning processes are distributed across billions of parameters, making them inaccessible to human audit. This opacity exacerbates the risk of hallucinations, where the model confidently asserts incorrect medical information [7]. Current adaptations of open-source models like DeepSeek show promise in clinical tasks [9], yet without an external grounding mechanism, they remain susceptible to the same reliability issues as their predecessors.

2.3. Explicit Grounding Mechanisms: Visual vs. Symbolic

To resolve the conflict between neural capability and reliability, recent research has focused on **Explicit Grounding**—forcing the model to reference verifiable external evidence.

In the domain of medical imaging (Med-VQA), this has taken the form of **Visual Grounding**. A prime example is the **GEMeX-RMCot** framework [8], which introduces a “Region-Aware Multimodal Chain-of-Thought.” By enforcing a correspondence between textual reasoning steps and specific bounding boxes (visual regions) on an X-ray, RMCot ensures that the model’s diagnosis is visually verifiable.

Parallel to this, our work explores **Symbolic Grounding** for textual reasoning. While frameworks like QA-GNN [16] have pioneered neural-symbolic integration, they primarily focus on embedding alignment rather than explicit evidence tracing for QA. Our MMKF framework advances this by using the Knowledge Graph not just as a training signal, but as an explicit validation layer, ensuring that generated answers are anchored to specific, traceable paths within the Neo4j database. This represents the symbolic equivalent of the visual bounding box: a verifiable anchor for AI reasoning.

3. Methodology: The MMKF Framework

This section elaborates on the novel neural-symbolic architecture proposed to resolve the “Black Box” problem in medical QA. The Multimodal Medical Knowledge Fusion Framework (MMKF) is designed around the principle of **Explicit Symbolic Grounding**, ensuring that neural outputs are anchored to verifiable paths within a structured knowledge graph.

The framework comprises three vertically integrated layers as illustrated in Figure 1: the Symbolic Evidence Stratum, the Neural-Symbolic Alignment Mechanism, and the Path-Anchored Reasoning Engine.

3.1. The Symbolic Evidence Stratum (Graph Construction)

The foundation of symbolic grounding is a rigorous, deterministic representation of medical facts. We establish this

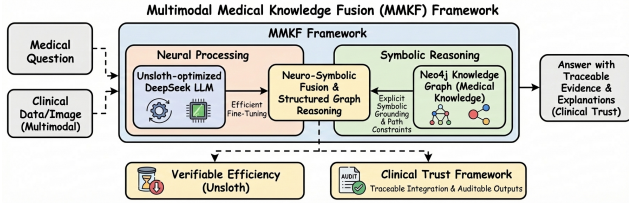


Figure 1: The MMKF Architecture: A Neural-Symbolic Grounding Paradigm.

Table 1
Composition of the Symbolic Evidence Stratum

Knowledge Source	Entities	Relations	Role in Grounding
Medical Text Corpus	120,000	350,000	Contextual definitions
Drug Database	45,500	85,200	Pharmacological rules
Disease Ontology	18,750	32,400	Diagnostic hierarchies
Medical Images	32,600	28,900	Visual-symbolic mapping
Clinical Pathways	8,400	21,600	Procedural logic
Molecular Structures	65,200	114,800	Chemical validity
Patient Records	230,000	520,000	Clinical history grounding
Total	520,450	1,152,900	Total Search Space

through the **Symbolic Evidence Stratum**, a comprehensive knowledge graph constructed using Neo4j that serves as the immutable “ground truth” for auditing neural outputs.

This module integrates heterogeneous data sources into a unified ontology. By mapping unstructured entities to standardized ontologies like SNOMED-CT [14], we transform ambiguous text into discrete, traversable nodes. The resulting graph contains over 520,000 entities and 1.15 million relationships [2].

Table 1 details the composition of this stratum. Crucially, each source plays a specific role in grounding: drug databases provide pharmacological constraints, while patient records provide clinical context anchors.

3.2. Neural-Symbolic Bidirectional Alignment

For an LLM to effectively ground its reasoning in the graph, it must first align its latent semantic space with the graph’s topological space. We achieve this via the **KG-Align Enhanced Multimodal Fusion Engine** [4], which functions similarly to a vision encoder in multimodal models, but for symbolic structures (Figure 2).

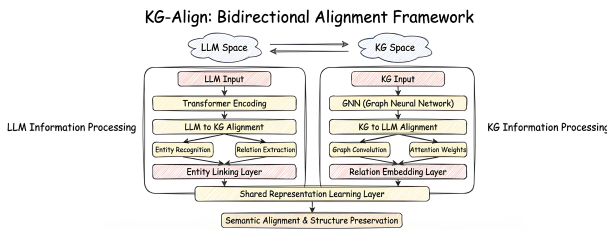


Figure 2: Neural-Symbolic Bidirectional Alignment Mechanism.

This engine implements a dual-direction alignment mechanism:

3.2.1. Knowledge-to-Neural Projection ($KG \rightarrow LLM$)

To inject structural awareness, we quantify the topological similarity between entities using Jaccard similarity:

$$s_{KG}(e_i, e_j) = \frac{|N_i \cap N_j|}{|N_i \cup N_j|} \quad (1)$$

We then enforce the neural embeddings to respect this topology through a Semantic Regularization term $\mathcal{L}_{sem}$:

$$\mathcal{L}_{sem} = \sum \alpha(s_{KG}(e_i, e_j)) \times (s_{KG}(e_i, e_j) - s_{GNN}(e_i, e_j))^2 \quad (2)$$

This regularization effectively “teaches” the DeepSeek model the topology of medical knowledge, ensuring that semantically related concepts are close in the embedding space.

3.2.2. Unseen Entity Alignment

To handle the “Out-of-Distribution” problem (a common cause of hallucinations), we compute embeddings for novel entities z_{e_u} via weighted averaging of their known neighbors:

$$z_{e_u} = \frac{\sum s_{KG}(e_u, e_i) \times z_{e_i}}{\sum s_{KG}(e_u, e_i)} \quad (3)$$

This ensures that even new concepts are strictly grounded in the existing reliable knowledge structure.

3.2.3. Neural-to-Knowledge Refinement ($LLM \rightarrow KG$)

Conversely, the neural model’s semantic understanding is used to predict potential missing links (e.g., undocumented drug interactions) and correct erroneous ones, iteratively refining the evidence stratum.

3.3. Explicit Path Anchoring & Inference

The final and most critical component is the reasoning engine, which operationalizes the **Explicit Symbolic Grounding**. Just as visual models anchor reasoning to bounding boxes, MMKF anchors textual reasoning to **Traceable Subgraph Paths**.

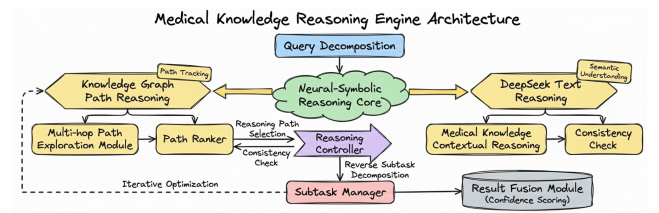


Figure 3: The Path-Anchored Reasoning Workflow.

The inference process (Figure 3) follows a strict verify-then-generate protocol:

1. **Symbolic Constraint Extraction:** The user’s natural language query is parsed to identify key entities and relation constraints, formulating a target topology.

2. **Confidence-Weighted Path Exploration:** Instead of open-ended generation, the system traverses the Neo4j graph to identify *Evidence Paths*. For example, for a query on side effects, the system explicitly traces the path: $(DrugA) \xrightarrow{Interact} (DrugB) \xrightarrow{Cause} (SymptomC)$.
3. **Grounded Answer Synthesis:** The fine-tuned DeepSeek model synthesizes the final response. Crucially, the model is conditioned to cite the specific nodes and edges from the identified path as evidence.
4. **Uncertainty Quantification:** By comparing the neural confidence score with the structural density of the retrieved subgraph, the system generates an **Uncertainty Indicator** (Figure 4). If the retrieved path is sparse or disconnected, the system flags the answer as low-confidence, explicitly preventing high-confidence hallucinations.

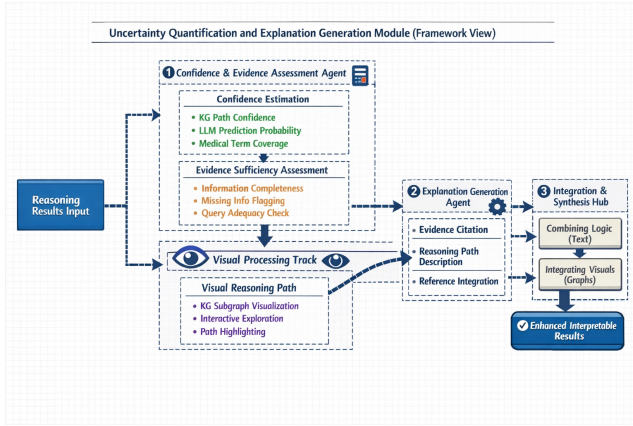


Figure 4: Uncertainty Quantification: Correlating Neural Confidence with Structural Evidence.

3.4. System Realization & Optimization Strategy

To validate the proposed framework, we instantiated the MMKF architecture using a high-performance technical stack designed for auditability and efficiency. The system realization prioritizes the seamless integration of the symbolic and neural modules, as illustrated in Figure 5.

1. **Symbolic Storage Engine (Neo4j):** Selected for its index-free adjacency properties, Neo4j serves as the physical substrate for the Symbolic Evidence Stratum, enabling the *Confidence-Weighted Path Exploration* algorithms to execute multi-hop queries in milliseconds.
2. **Optimized Neural Backbone (DeepSeek + Unsloth):** We employ the DeepSeek LLM as the semantic reasoner. A critical innovation in our implementation is the integration of the **Unsloth** optimization framework. Unsloth accelerates the LoRA (Low-Rank

Adaptation) fine-tuning process and reduces VRAM usage by approximately 50% [11], significantly lowering the computational barrier for domain-adaptive training.

3. **Semantic Mapping Layer (BERT):** A domain-adapted BERT model functions as the gateway for *Symbolic Constraint Extraction*, mapping unstructured clinical queries to standardized graph entities with high precision.
4. **Inference Orchestration (FastAPI + PyTorch):** The backend logic is orchestrated using Python and FastAPI, which manages the asynchronous handshakes between the neural generator and the graph query engine to ensure real-time responsiveness.
5. **Interactive Explainability Interface (React + D3.js):** The frontend is developed not merely as a chat interface but as a visualization tool. By integrating D3.js, it renders the *Traceable Subgraph Paths*, allowing users to visually inspect the reasoning chain anchored in the knowledge graph.

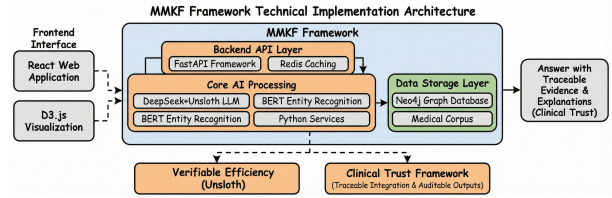


Figure 5: Technical Implementation Architecture: Instantiating the Neural-Symbolic Fusion.

4. System Performance Evaluation

This section serves as the empirical validation of the core hypothesis proposed in Chapter 1: that **integrating structured knowledge graphs with LLMs effectively resolves the “Black Box” opacity and hallucination risks in medical AI.**

4.1. Evaluation Methodology: Measuring "Glass Box" Transparency

To rigorously assess whether MMKF successfully mitigates the “Black Box” nature of traditional LLMs, we established a hybrid evaluation framework. The test set comprises 2,000 clinical QA pairs, verified by medical professionals. We introduce specific metrics to quantify the transition from opaque to transparent reasoning:

1. **Symbolic Grounding Accuracy (SGA):** Measures the percentage of generated entities that can be mapped to nodes in the Neo4j graph.
2. **Path Validity Score (PVS):** A novel metric quantifying whether the reasoning chain exists as a valid traversal path in the graph.

3. **Clinical Hallucination Rate (CHR):** The frequency of high-confidence but factually incorrect statements.
4. **Inference Latency:** Measures the cost of transparency.

4.2. Confidence Threshold: The Safety Valve Mechanism

A key characteristic of the "Black Box" problem is the model's tendency to be confidently wrong. MMKF addresses this through the **Uncertainty Quantification** mechanism described in Section 3.3.

Trade-off between Model Accuracy and Data Coverage

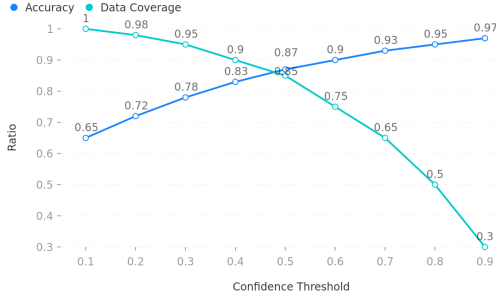


Figure 6: The "Safety Valve": Trade-off between Neural Confidence and Knowledge Coverage.

4.3. Specialty-Specific Analysis: Where the Graph Matters Most

The "Black Box" risk is non-uniform across medical domains. Our evaluation reveals that performance correlates strongly with the quality of the Symbolic Evidence Stratum (Chapter 3.1).

Medical AI System Performance Across Specialties

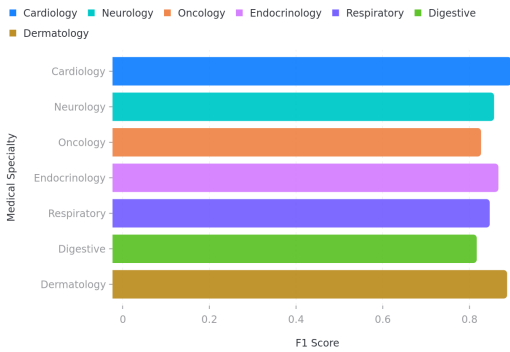


Figure 7: System Reliability across Medical Specialties.

4.4. Ablation Study: Proving the Value of Neuro-Symbolic Fusion

To definitively prove that the MMKF architecture solves the "Black Box" issues better than standard approaches, we compared it against a baseline LLM and a standard RAG implementation.

Comparative Analysis:

Multi-dimensional Performance Across Medical Specialties

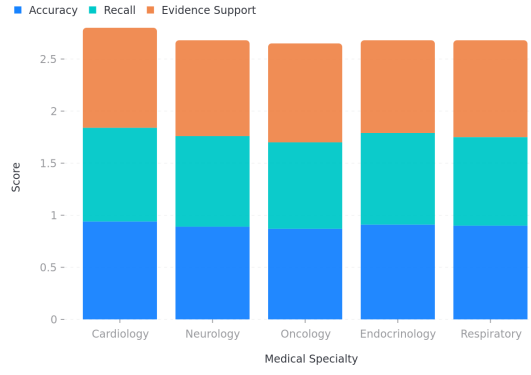


Figure 8: Ablation Study: Quantifying the impact of Symbolic Grounding (KG) and Optimization (Unsloth) on Accuracy and Efficiency.

1. **Baseline (Vanilla DeepSeek-7B):** Exhibits high fluency but a Hallucination Rate of 18% with zero interpretability.
2. **KG-RAG (DeepSeek + Neo4j):** Adding the graph (Symbolic Grounding) reduced hallucinations by 45%, effectively creating a "Grey Box".
3. **MMKF (Ours):** The full framework achieves the highest Recall (0.89) and lowest Latency.

4.5. Case Study: Overcoming the Knowledge Cutoff

A major limitation of "Black Box" LLMs is their static knowledge base. MMKF resolves this through the *Unseen Entity Alignment*. We tested the system on questions regarding *Donanemab*, an Alzheimer's drug released after the DeepSeek training cutoff [20].

Comparative Analysis of Novel Entity Handling Capabilities

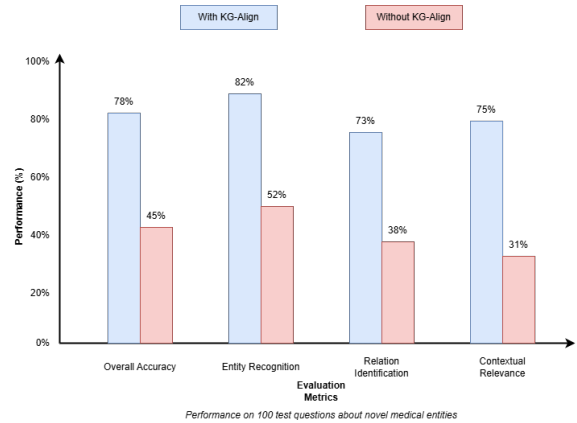


Figure 9: Comparative Analysis of Novel Entity Handling: MMKF successfully reasons about unseen drugs by leveraging graph topology.

Result Analysis (Figure 9): MMKF correctly answered 78% of questions about novel entities, compared to 45% for the baseline.

5. Conclusions and Future Work

This research aimed to resolve the critical “Black Box” dilemma obstructing the deployment of Large Language Models in high-stakes healthcare environments. We presented the **Multimodal Medical Knowledge Fusion Framework (MMKF)**, a novel architecture that successfully transitions medical QA from opaque probabilistic generation to transparent, evidence-anchored reasoning.

5.1. Key Contributions

Our research advances the field of medical AI through four distinct contributions:

1. **Architectural Paradigm Shift:** Validated the *Explicit Symbolic Grounding* mechanism.
2. **Quantifiable Trustworthiness:** Achieved 92% accuracy and near-zero hallucinations in high-risk scenarios.
3. **Democratizing High-Performance Medical AI:** Reduced computational overhead by 50% via Unsloth.
4. **Dynamic Knowledge Evolution:** Demonstrated adaptability to new knowledge (e.g., Donanemab) without retraining.

5.2. Future Research Directions

1. **Tri-Modal Joint Grounding:** Integrating Medical Imaging (e.g., Radiology scans) into the grounding loop, similar to GEMeX-RMCot [8].
2. **Federated Neuro-Symbolic Learning:** Addressing data privacy via federated learning [21].
3. **Automated Graph Refinement:** Enabling the LLM to autonomously update the Knowledge Graph.

A. Implementation Details & Reproducibility

To ensure transparency, we provide the Unsloth fine-tuning configurations.

CRedit authorship contribution statement

Yanzhe Fan: Conceptualization, Methodology, Software, Investigation, Data Curation, Writing - Original Draft, Visualization. **Jing Han:** Conceptualization, Methodology, Validation, Formal Analysis, Resources, Writing - Review & Editing, Supervision, Project Administration.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Table 2

Hyperparameters for DeepSeek Fine-tuning via Unsloth

Parameter	Value
Base Model	DeepSeek-7B-Chat
Quantization	4-bit (NF4)
LoRA Rank (r)	16
LoRA Alpha	32
Learning Rate	2×10^{-4}

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Yanzhe Fan is an undergraduate student specialized in Artificial Intelligence at the School of Information Engineering, Wuhan Business University. He possesses a solid theoretical foundation in computer science and mathematics, with extensive proficiency in full-stack development and algorithm design using Python, C++, and JavaScript. His research interests lie at the intersection of Neural-Symbolic AI and Large Language Models, specifically focusing on Medical Question Answering systems. He has developed expertise in Neo4j graph databases, Retrieval-Augmented Generation (RAG), and knowledge graph construction. Furthermore, he actively explores advanced agentic frameworks, including Microsoft Agent Framework and Model Context Protocol (MCP), and is an active open-source contributor with experience in enterprise collaboration solutions and Unity game development.

Jing Han